

Jorge Ortega Camazón

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EDUCATION:

Northeastern University, Boston, MA **May 2026**
Master of Science in Robotics, Concentration: Electrical and Computer Engineering **GPA: 4.0**
Relevant Courses: Robotics Sensing and Navigation, Reinforcement Learning, Robot Mechanics and Control
Universidad de León, León, Spain **Jul 2024**
Bachelor of Science in Industrial Electronic and Automation Engineering
Relevant Courses: Robotic Prototypes, Automatic Control, Power Electronics, Digital Electronics
*Studied abroad during the academic years 2018-2019 and 2022-2023 in USA and Italy respectively

TECHNICAL SKILLS:

Software: LTspice, MATLAB, EAGLE, Fusion 360, SolidWorks, AutoCAD, ROS, Gazebo, RViz, MS Office Tools
Hardware: Electromechanical Actuators, Soldering, PCB design, Microcontrollers, Electronic equipment
Programming Languages: Python, C++, MATLAB
Robotics and Automation: Impedance Control, QP Algorithms, SLAM, LiDAR, IMU, GPS, MuJoCo
Languages: Spanish (Native), English (Fluent), Italian (Conversational)

RELEVANT EXPERIENCE:

TITAN Advanced Energy Solutions, Salem, MA **January 2026 – August 2026**
Ultrasound and Battery Science Intern

- Engineered a standardized fixture-swapping and spatial alignment protocol for cylindrical cells, reducing tank setup time by 50% (from 8 to 4 hours) while improving Time of Flight alignment accuracy from $<1\mu s$ to $\sim 250ns$
- Characterized 70 air-coupled transducers, analyzing time/frequency domain features. Developed an interactive data visualization tool to compare acoustic waveforms, integrating a weighted Blossom algorithm to automatically compute the optimal transmitter-receiver pairs based on user-defined performance metrics
- Engineered a Python simulation tool to optimize the spatial configuration of a 5 sensor-pair inspection array on a cross-axis gantry system for real-time electrode manufacturing. Implementing an optimization algorithm to maximize scan coverage and minimize maximum missed defects
- Developed a baseline calibration script utilizing a known-density prismatic block to evaluate system drift and restore highly precise transducer alignment, establishing calibration benchmarks for cell cycling campaigns

Graduate Researcher **May 2025 – Present**
Silicon Synapse Lab, Northeastern University

- Implemented multi-language (Python/C++/MATLAB) workflows using Pinocchio library to import biped robot URDFs, compute inverse/forward dynamics and kinematics, and visualize inertial properties in Drake
- Assembled and characterized two electromechanical actuators with harmonic drive, identifying and resolving friction-related performance issues, and subsequently integrated them into a full robotic leg for impedance control research on the biped robot platform Harpy

PROJECTS:

Stable Walking on Biped Robot using RL **Feb – Apr 2025**

- Trained a customized biped robot in the MuJoCo Walker2D-v5 environment using a PPO algorithm based on the Stable Baselines3 library, achieving stable and consistent forward locomotion
- Developed tailored Python code enabling parallel training across 4 environments, integrating TensorBoard for real-time performance tracking and facilitating hyperparameter tuning
- Modified reward functions, torque limits and agent's geometry to encourage realistic motion dynamics, enabling energy-efficient walking and high-speed running behavior across tests

Synchronization and Coordination of Mobile Robots **Nov – Dec 2024**

- Simulated a multi agent robot system for autonomous exploration within a shared environment based on the open-source *TurtleBot 3* robot ROS packages
- Modified autonomous navigation algorithms and configured SLAM launch files for 3 individual robots to enable simultaneous multi-agent exploration in customized Gazebo environment using ROS Noetic
- Configured Rviz for real-time visualization of each robot's exploration process and dynamically merged maps to generate a complete representation of the environment

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Control System Optimization for a Two-Elevator Building

Nov – Dec 2024

- Developed and simulated three elevator control systems in MATLAB, utilizing parallel computing techniques to optimize time and energy efficiency, in real-time multi-elevator operation and dynamic request allocation
- Evaluated control system performances by assessing efficiency metrics such as total travel time, floors traversed and workload balance, achieving a 250% improvement in time and optimal workload balance

Low-Cost, Multifunctional Autonomous Robot

Feb – Jul 2024

- Engineered an autonomous robot for indoor mapping with a focus on cost-efficiency using durable, lightweight materials, optimizing weight distribution for stable operation in diverse environments
- Integrated an RPLiDAR A1M8 for real-time mapping and a GPS NEO-M8N for localization, enabling precise navigation with ESP32 microcontrollers
- Developed custom firmware to process and filter data, manage communication between microcontrollers and sensors, and implement motor control and navigation logic.
- Achieved a functional prototype capable of autonomous navigation and mapping with a cost inferior to \$300, demonstrating a potential affordable alternative to commercial Autonomous Mobile Robots